CMPG 311 Project (Phase 2)
Liftex Delivery Intelligence & Attendance System
GROUP 7

GROUP INFORMATION

Name and Surname	Student Number	Contact
Lehlogonolo Mogola	50795945	50795945@mynwu.ac.za
Koketso Mengwai	44419465	44419465@mynwu.ac.za
Surprise Selowa	45192472	45192472@mynwu.ac.za
Sandisile Nkosi	51768844	51768844@mynwu.ac.za
Rabetswe Modikwe	45914494	45914494@mynwu.ac.za
Olebogeng Mokobane	31120997	31120997@mynwu.ac.za
Potako Makofane	43436625	43436625@mynwu.ac.za

1.1 Business Rules

1. Employee & Role Management

- The system must store employee information before they can participate in operations.
- Each employee is assigned exactly one role.
- Roles include Driver, Fleet Admin, Finance Admin, and Legal Admin.
- A role can be assigned to multiple employees.

2. Warehouse Management

- Each employee must be assigned to one home warehouse
- A warehouse can employ multiple employees.

3. Attendance Management

- Employees must check in at the beginning of their shift and check out at the end.
- The warehouse ID's where the employee checked in and out are recorded, including when they checked in.
- An employee must check in and check out at a Warehouse and it is possible for an employee to check in and out at the same warehouse.

4. Delivery Shift Management

- Each delivery shift is assigned to one employee.
- An employee can have multiple delivery shifts.
- Each shift must be associated with one vehicle.
- A vehicle can be used in multiple shifts.

5. Parcel & Delivery Management

- Each employee has a daily delivery shift
- -Employees can work up to one delivery shift per day but multiple shifts over a period of time. Each delivery shift is associated with only one employee.

- A delivery shift can have multiple parcel assignments.
- Each Parcel will have a corresponding status from (“Pending”, “In-Transit” and “Delivered”).
- A parcel is stored in one warehouse.

6. Payroll Management

- Each employee can have multiple payroll records recorded monthly.
- A payroll record belongs to one employee.

7. Audit Logging

- Every system action (INSERT, UPDATE, DELETE) must be recorded.
- Each audit log entry must be linked to an employee.
- The audit log must store details of the action performed and timestamp.

8.Fuel Transactions

- Each delivery Shift can have multiple fuel transaction records, but each fuel transaction belongs to a single delivery shift

9. Theft/Fault management

- -Employee Fuel usage is monitored for total fuel consumption as well as missing fuel not accounted for by the CAN bus port
- -Upon detection of missing fuel, fleet admins will conduct an inspection of the vehicle as well as a hearing to determine whether the missing fuel status is(“Stolen” or “Mechanical Fault”). If theft is confirmed, then the missing fuel cost is deducted from the employee’s monthly pay and added as part of their total consumed fuel to be considered when determining if they are applicable for a “Fuel Bonus”

10. System Constraints

- A delivery shift cannot exist without an assigned employee and vehicle.
- A parcel cannot be delivered without being assigned to a delivery shift.
- An employee’s total parcel load in Kilograms must not exceed their assigned vehicle’s maximum payload

1.2 Relationships Between Our Tables Descriptions

Role ↔ Employee: One-to-Many (1:M). Each employee has one role; a role can have many employees. **Weak** relationship.

Warehouse ↔ Employee: One-to-Many (1:M). Each employee belongs to one warehouse; a warehouse can employ many employees. **Weak** relationship.

Employee ↔ Delivery Shift: One-to-Many (1:M). Each shift belongs to one employee, but an employee can work more than one delivery shift. **Weak** relationship.

Vehicle ↔ Delivery Shift: One-to-Many (1:M). Each shift uses one vehicle, and each vehicle can be used in multiple delivery shifts. **Weak** relationship.

Delivery Shift ↔ Parcel: One-to-Many (1:M). Each delivery Shift can deliver multiple parcels, but each parcel belongs to one delivery shift. This is a **weak** relationship.

Parcel ↔ Status: One-to-Many (1:M). Each Parcel has only one status but a status can apply to many parcels. **Weak** relationship.

Employee ↔ Payroll Record: One-to-Many (1:M). Each payroll record belongs to one employee but an employee can have multiple payroll records. This is a **weak** relationship.

Employee ↔ Audit Log: One-to-Many (1:M). Each audit log belongs to one employee, and an employee can generate multiple audit logs. The audit log entity was purposely given a composite primary key, containing the primary key of the **Employee** table to give it a **strong** relation with **Employee** in order to meet the rubric requirements.

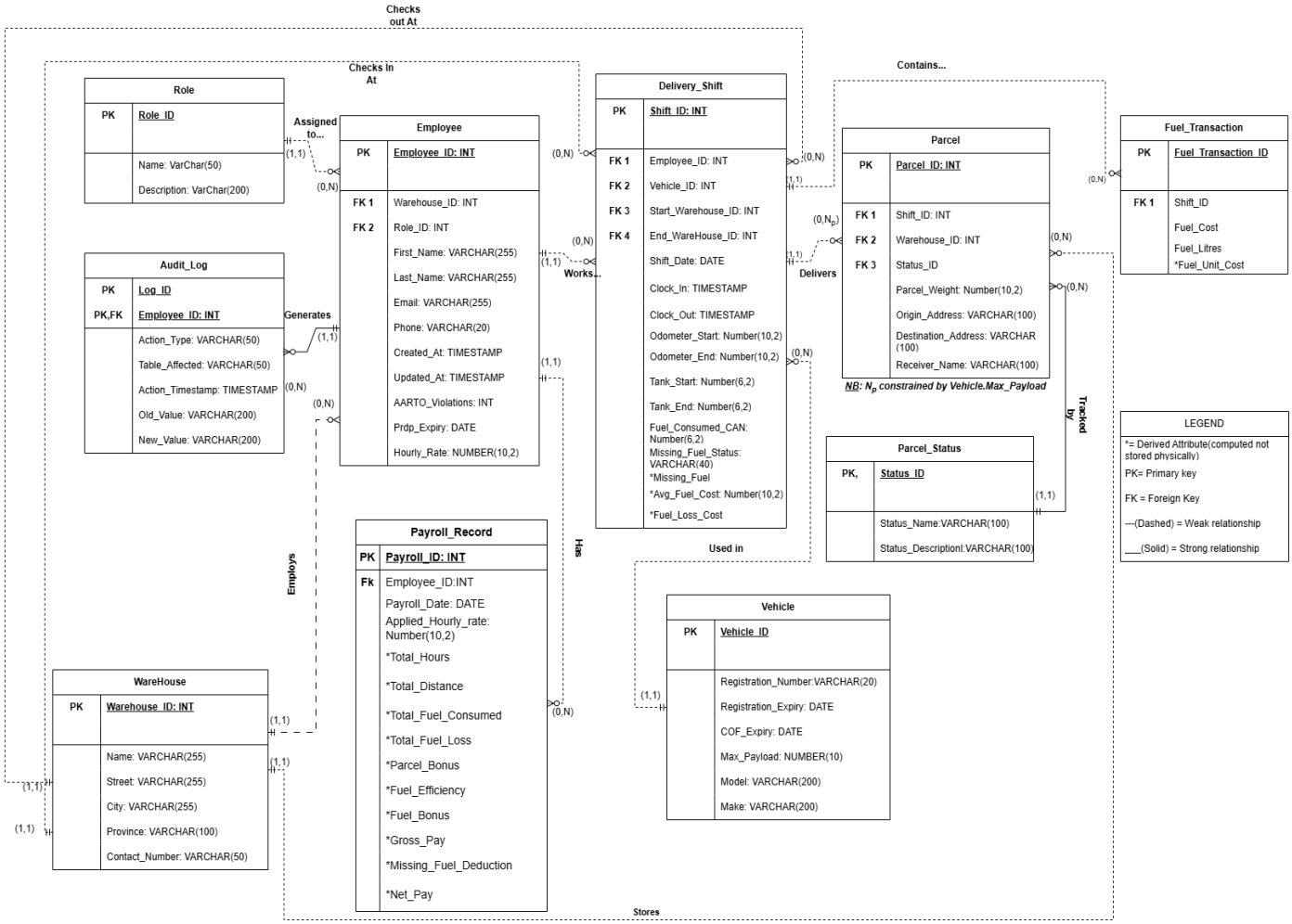
Warehouse ↔ Parcel: One-to-Many (1:M). Each Warehouse may store multiple parcels, but a parcel is originally stored and picked up from one warehouse. **Weak** relationship.

Delivery Shift ↔ Fuel Transaction: One-to-Many (1:M). Each shift may contain multiple fuel transactions but each fuel transaction belongs only to a single shift. **Weak** relationship.

Delivery Shift ↔ Warehouse: One-to-Many (1:M). A single warehouse can be the starting or ending point for many shifts, but each shift has exactly one starting and ending warehouse. **Weak** relationship.

Apart from **Employee** and **Audit_Log,** all other relationships are defned as **weak** because the child entity does not possess, as part of its primary key, a primary key from the parent entity

1.2 ERD (Entity Relationship Diagram)



1.4 Derived Attributes

Fuel Transaction Table

- **Fuel_Unit_Cost** is derived from **(Fuel_Cost and Fuel_Litres)**

Delivery Shift Table

- **Missing_Fuel** is derived from **Tank_Start, Tank_End, Fuel_Consumed_CAN**(fuel consumed as measured by the CAN bus port), and the sum of all **Fuel_Litres** (from **FUEL_TRANSACTION)** associated with each delivery shift.

- **Avg_Fuel_Cost** is derived from the average of all **Fuel_Unit_Cost** across each **Fuel_Transaction** entity associated with a single delivery shift.

- **Fuel_Loss_Cost** is derived from the total cost of all **Missing_Fuel** during a shift.

Payroll Record Table

- **Total_hours** is derived from total (**Clock_In and Clock_Out**) across all **Delivery_Shift** entities in the relevant month corresponding to the relevant **Employee_ID**

- **Total_Fuel_Consumed** is derived from the sum of all **Fuel_Consumed_CAN** for all delivery_Shift entities in the relevant month as well as all **Missing_Fuel** flagged as "stolen". The delivery shift entities must correspond to the relevant **Employee_ID**

- **Total_Fuel_Loss** is derived from total **Missing_fuel** across each **Delivery_Shift** entity in the relevant month where **Missing_Fuel_Status** was flagged as "Stolen" and corresponding to the relevant **Employee_ID**

- **Total_Distance** is derived from the sum the difference(**Odometer_Start - Odometer End)** across each **Delivery_Shift** entitiy in a month corresponding to the relevant **Employee_ID**

- **Parcel_Bonus** is derived from total **Parcels** associated with each driver with a "Delivered" status across multiple **Delivery_Shift** entities in the relavant month corresponding to the relevant **Employee_ID**.

- **Fuel_Efficiency** is derived from (**Total_Fuel_Consumed & Total_Distance)** from the relavant employee's **Payroll_Record** for the relevant month.

- **Fuel_Bonus** is derived by comparing the driver's monthly fuel efficiency (**Total_Fuel_Consumed/Total Distance)** and comparing it to the company's average fleet efficiency (The average of all Fuel_Efficiency from all the company's employee's **Payroll_Record** entities in that relevant month.

- **Gross_Pay** is derived from **Total_Hours** and **Applied_Hourly_Rate**

- **Missing_Fuel_Deduction** is derived from the total of all fuel_Loss_Cost across all **Delivery_Shift** entities in the relevant month where **Missing_Fuel** has a "Stolen" status, and corresponding to the relevant **Employee_ID.**

- **Net_Pay,** for a specific employee's **Payroll_Record,** is derived from **Gross_Pay, Parcel_Bonus, Fuel_Bonus, and Missing_Fuel_Deduction.**

1.5 Logical Design

Warehouse (<u>Warehouse_ID</u>, Name, Street, City, Province, Contact_Number)

PRIMARY KEY: Warehouse_ID

Role (<u>Role_ID</u>, Name, Description)

PRIMARY KEY: Role_ID

Vehicle(<u>Vehicle_ID</u>, Registration_Number, Registration_Expiry, COF_Expiry, Max_Payload, Model, Make)

PRIMARY KEY: Vehicle_ID

Parcel_Status(<u>Status_ID</u>, Status_Name, Status_Description)

PRIMARY KEY: Status_ID

Payroll_Record(<u>Payroll_ID</u>, *Employee_ID*, Payroll_Date, Applied_Hourly_Rate)

PRIMARY KEY: Payroll_ID

FOREIGN KEY: Employee_ID REFERENCES Employee

Fuel_Transaction (Fuel_Transaction_ID, *Shift_ID*, Fuel_Cost, Fuel_Litres)

PRIMARY KEY: Fuel_Transaction_ID

FOREIGN KEY: Shift_ID REFERENCES Delivery_Shift

Audit_Log(<u>Log_ID, *Employee_ID*</u>, Action_Type, Table_Affected, Action_Timestamp, Old_Value, New_Value)

PRIMARY KEY: Log_ID, Employee_ID

FOREIGN KEY: Employee_ID REFERENCES Employee

Employee (<u>Employee_ID</u>, *Warehouse_ID, Role_ID*, First_Name, Last_Name, Email, Phone, Created_At, Updated_At, AARTO_Violations, Prdp_Expiry, Hourly_Rate)

PRIMARY KEY: Employee_ID

FOREIGN KEY: Warehouse_ID REFERENCES Warehouse

Role_ID REFERENCES Role

Delivery_Shift(<u>Shift_ID</u>, *Employee_ID, Vehicle_ID, Start_Warehouse_ID, End_Warehouse_ID,* Shift_Date, Clock_In, Clock_Out, Odometer_Start, Odometer_End, Tank_Start, Tank_End, Fuel_Consumed_CAN, Missing_Fuel_Status)

PRIMARY KEY: Shift_ID

FOREIGN KEY: Employee_ID REFERENCES Employee

Vehicle_ID REFERENCES Vehicle

Start_Warehouse_ID REFERENCES Warehouse

End_Warehouse_ID REFERENCES Warehouse

CONSRAINTS: Employee_ID and Vehicle_ID must not be null.

Parcel(<u>Parcel_ID</u>, *Shift_ID, Warehouse_ID, Status_ID,* Parcel_Weight, Origin_Address, Destination_Address, Receiver_Name)

PRIMARY KEY: Parcel_ID

FOREIGN KEY: Shift_ID REFERENCES Delivery_Shift

Warehouse_ID REFERENCES Warehouse

Status_ID REFERENCES Status